

GPS → Mitsuba Coordinate Transformation

Sionna 0.19.2 · Nottingham Ray-Tracing Simulation

Overview

Every GPS coordinate from the Ofcom drive-test CSV must be converted into a 3-D position (x, y, z) in Mitsuba's local scene space before it can be used as a receiver or transmitter. The transformation has three stages:

Stage	Input	Output	Tool
1 — GPS → UTM	lon/lat (degrees)	easting/northing (m, absolute)	pyproj EPSG:4326→32630
2 — UTM → Local	easting/northing (m)	x/y (m, scene-centred)	subtract bbox centre
3 — Z from mesh	local x/y (m)	z (m, DSM surface height)	Mitsuba ray_cast

Stage 1 — WGS84 (GPS) → UTM Zone 30N

Input: longitude/latitude in decimal degrees (EPSG:4326, WGS84) — the raw format stored in *nottingham3602.csv*.

Output: easting (X) and northing (Y) in absolute metres, UTM Zone 30N (EPSG:32630). At UK latitude, 1° longitude ≈ 67 km and 1° latitude ≈ 111 km.

```
from pyproj import Transformer
gps_to_utm = Transformer.from_crs("EPSG:4326",
    "EPSG:32630", always_xy=True) # Example — TX site lon, lat = -1.2559, 52.9863
easting, northing = gps_to_utm.transform(lon, lat) # → easting = 457,842 m # → northing = 5,873,415 m
```

`always_xy=True` forces (longitude, latitude) order regardless of CRS axis convention — without this pyproj may silently swap the axes.

Stage 2 — UTM Absolute → Scene-Local (centred at 0, 0)

Mitsuba does not know about geographic coordinates. The scene PLY meshes are built relative to the **centre of the scene bounding box**, so all positions must use the same origin.

Origin computation (done once at scene build time)

```
# Scene bbox midpoint in WGS84
center_lon = (SCENE_WEST + SCENE_EAST) / 2 # e.g. -1.327
center_lat = (SCENE_SOUTH + SCENE_NORTH) / 2 # e.g. 52.960
# Convert to UTM to get the metric origin
utm_center_x, utm_center_y = gps_to_utm.transform(center_lon, center_lat)
utm_center_x ≈ 453,475 m # utm_center_y ≈ 5,870,463 m
```

Per-point conversion

```
def gps_to_local(lon, lat):
    ux, uy = gps_to_utm.transform(lon, lat)
    local_x = ux - utm_center_x # metres east of scene centre
    local_y = uy - utm_center_y # metres north of scene centre
    return local_x, local_y
# Example — drive-test point
local_x, local_y = gps_to_local(-1.2508, 52.9875)
# → local_x = +4,711 m (east of scene centre) # → local_y = +3,074 m (north of scene centre)
```

The TX at (-1.2559, 52.9863) maps to approximately (4711 m, 3074 m) in local coordinates — it is near the north-east edge of the scene, not at the origin.

Stage 3 — Z Height from Mitsuba Scene Mesh (ray_cast)

The terrain mesh was built from the AWS DSM (Digital Surface Model — includes building rooftops and canopy). To place a receiver or transmitter consistently with this mesh, a downward ray is fired from $z = 2000$ m and the intersection height is returned.

```
import mitsuba as mi def ray_cast_ground_z(x, y, max_height=2000.0): """Return scene-local
Z of ground/surface at (x, y).""" ray = mi.Ray3f( mi.Point3f(float(x), float(y),
max_height), # origin mi.Vector3f(0.0, 0.0, -1.0) # direction: straight down ) si =
scene.mi_scene.ray_intersect(ray) if bool(si.is_valid()): return float(si.p.z) # Z of
first surface hit return 0.0 # fallback if no intersection # Example ground_z =
ray_cast_ground_z(4711.0, 3074.0) # → ground_z = 67.1 m (DSM surface = terrain + any
building) # Final receiver position rx_z = ground_z + 1.5 # 1.5 m antenna height AGL # →
rx_z = 68.6 m # Final transmitter position tx_z = ground_z + 17.0 # 17 m mast height AGL #
→ tx_z = 84.1 m
```

Why ray_cast and not DEM tiles? The DTM (bare-earth) tiles give ~65 m at the TX location while the DSM terrain mesh reads ~131 m ASL there (67 m local). Using DTM would place the TX 49 m underground. ray_cast is always consistent with the loaded scene geometry.

Complete Example — One Drive-Test Point

Step	Operation	Value
CSV input	Raw Ofcom measurement	lon=-1.2508, lat=52.9875, RSSI=-46.2 dBm
Stage 1	GPS → UTM 30N	easting=458,186 m northing=5,873,537 m
Stage 2	UTM → local (subtract centre)	local_x=+4,711 m local_y=+3,074 m
Stage 3	ray_cast Z at (4711, 3074)	ground_z=67.1 m
RX height	ground_z + 1.5 m antenna AGL	rx_z = 68.6 m
Mitsuba pos.	Receiver position vector	[4711.0, 3074.0, 68.6] metres

Reference Frames Summary

Frame	EPSG	Origin	Units	Used for
WGS84	4326	Earth centre	degrees	CSV input, GPS devices
UTM 30N	32630	Zone SW corner	metres abs	intermediate (pyproj)
Scene-local	—	Scene bbox centre	metres rel	Mitsuba, PLY meshes, all outputs
Mitsuba Z	—	Scene origin plane	metres	DSM surface from ray_cast

DSM vs DTM — Why It Matters

The AWS elevation-tiles-prod bucket provides **DSM** (Digital Surface Model): elevation includes buildings, trees, and other above-ground objects. The scene terrain mesh is built from DSM tiles.

Model	What it measures	At TX site (Nottingham)	Effect if used for Z
DSM	Surface (buildings + terrain)	~131 m ASL / 67 m local	TX at correct height ✓
DTM	Bare earth only	~65 m ASL / 1 m local	TX 49 m underground ✗

Solution: always use `ray_cast_ground_z()` — it fires a ray into the loaded Mitsuba scene (built from DSM) so the returned Z is guaranteed to match the scene geometry, regardless of which elevation dataset was used.